



MIRPUR UNIVERSITY OF SCIENCE AND TECHNOLOGY

(MUST)

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DEPARTMENT:	SOFTWARE ENGINEERING

SOFTWARE QUALITY ENGINEERING - LAB SESSION 3

STLC & Test Documentation Setup

1. Introduction

This report presents the completed work for Software Quality Engineering Lab Session 3: STLC & Test Documentation Setup. The selected project is the University Library Management System (LMS), a web-based system used to manage authentication, book searching, loan management, and notification activities for students and library staff.

The lab work includes Jira test management setup, Requirement Traceability Matrix (RTM), IEEE 829 based Test Plan, and Jira test case documentation. The purpose is to show that testing activities are organized professionally before formal test execution begins.

2. Lab Objectives

- Configure a Jira Scrum project for test management.
- Prepare a Requirement Traceability Matrix for all 16 LMS requirements.
- Write an IEEE 829 based Test Plan for the LMS.
- Create reusable test cases for Jira execution.
- Provide a clean submission-ready single Word report.

3. Jira Project Configuration Summary

Item	Details
Project Name	SQE-Lab-Sulman-Farooq
Project Type	Scrum Project
Board Columns	To Do In Progress Testing Done
Issue Types	Story/Test Case, Bug, Task, Sub-task
Custom Fields	Test Priority, Test Type, STLC Phase
Sprint Name	Week 3 Lab
Jira Project URL	Paste your Jira project URL here before final submission

Note: The Jira URL and screenshots must be verified in the actual Jira Cloud workspace before submission. The project should remain private, but the instructor should be added as Viewer.

4. Requirement Traceability Matrix (RTM) Summary

The RTM maps each LMS requirement to one or more test cases. It confirms that all 16 requirements are covered and no requirement is left as a testing gap.

Metric	Value
Total Requirements	16
Covered Requirements	16
Gap Requirements	0
Coverage Percentage	100%

4.1 Important Final Actions Before Upload

- Open your Jira Cloud project and confirm Sprint 1 is active.
- Paste your actual Jira project URL in Section 3 of this file.
- Add the instructor as Viewer in Project Settings > People.
- Attach or show Jira screenshots separately if your teacher asks for visual proof.
- Submit this Word file as the single combined report.

5. Complete Requirement Traceability Matrix

Req. ID	Requirement	Priority	TC ID(s)	Test Case Title	Test Type	Covered? (Y/N/Gap)	Tester
REQ-01	User shall log in with valid credentials	High	TC-001, TC-002	Valid login / Invalid username login	Functional	Y	student
REQ-02	System shall lock account after 3 failed attempts	High	TC-003	Account lockout after 3 failed attempts	Functional, Security	Y	student
REQ-03	Password reset email within 2 minutes	Medium	TC-004	Password reset email delivery	Functional, Performance	Y	student
REQ-04	Session expires after 30 min inactivity	High	TC-005, TC-006	Session timeout / Session timer reset on activity	Functional, Security	Y	student
REQ-05	Search by title, author, ISBN	High	TC-007, TC-008, TC-009	Search by title / by author / by ISBN	Functional	Y	student
REQ-06	Filter by genre and availability	Medium	TC-010, TC-011	Filter by genre / Filter by availability	Functional	Y	Sulman
REQ-07	Search results display within 2 seconds	High	TC-012	Search response time under 2 seconds	Performance	Y	Sulman
REQ-08	Pagination for results > 20	Low	TC-013	Pagination with 21+ results	Functional	Y	Sulman
REQ-09	Issue book loan to registered student	High	TC-014, TC-015	Issue loan valid student / unregistered student	Functional	Y	Sulman
REQ-10	Set due date to 14 days from issue	High	TC-016, TC-017	Due date calculation standard / month boundary	Functional	Y	Sulman
REQ-11	Renew loan if no hold requests	Medium	TC-018, TC-019	Renew loan no hold / hold exists	Functional	Y	Sulman
REQ-12	Calculate overdue fines	Medium	TC-020, TC-021	Fine calculation 1 day / 10 days overdue	Functional	Y	Sulman
REQ-13	Email reminder 2 days before due date	Medium	TC-022	Email reminder sent 48 hours before due	Functional	Y	Sulman
REQ-14	SMS alert on overdue items	Low	TC-023	SMS alert on overdue loan	Functional	Y	Sulman
REQ-15	User configures notification preferences	Low	TC-024, TC-025	Enable email notification / Disable SMS notification	Functional, Usability	Y	Sulman
REQ-16	System emails dispatched within 5 minutes	Medium	TC-026	Email dispatch time under 5 minutes	Performance	Y	Sulman

6. IEEE 829 Test Plan

6.1 Test Plan Identifier

Field	Description
Test Plan ID	LMS-TP-001
Version	1.0
Date	5 May 2026
Project	University Library Management System
Prepared By	Sulman Farooq - SQE Lab Student
Reviewed By	Sir Asim Javaid SHB
Status	Draft - Pending Review

6.2 Introduction

The University Library Management System (LMS) is a web-based application developed for a university to manage book loans, returns, student accounts, and library staff operations. The system supports Students and Library Staff through a browser-based interface backed by a PostgreSQL database and Node.js server.

This test plan covers system-level testing of LMS Version 1.0, mainly Modules M1 Authentication and M2 Book Search for the current lab cycle. Remaining modules are documented in the RTM and will be executed in future testing cycles.

6.3 Test Items

Item	Version	Description
LMS Frontend (React.js)	v1.0.0	Browser-rendered React application served from staging environment.
LMS Backend (Node.js/Express)	v1.0.0	REST API server used by the LMS frontend.
LMS Database (PostgreSQL)	v14.2	Database containing books, loans, users, and notification data.
Authentication Service	v1.0.0	JWT-based login, session, and password reset service.
Test Data Set	LMS-TESTDATA-001	Pre-populated data with books, student accounts, and staff accounts.

6.4 Features to be Tested

Feature	Requirement	Priority
User login with valid credentials	REQ-01	High
Invalid login attempt rejection	REQ-01	High
Account lockout after exactly 3 failed attempts	REQ-02	High

Password reset email delivery within 2 minutes	REQ-03	Medium
Session expiry after 30 minutes of inactivity	REQ-04	High
Search by title, author, and ISBN	REQ-05	High
Filter by genre and availability	REQ-06	Medium
Search response under 2 seconds	REQ-07	High
Pagination for more than 20 results	REQ-08	Low

6.5 Features Not to be Tested

Feature Not Tested	Justification
M3 Loan Management (REQ-09 to REQ-12)	Out of scope for current test cycle; planned for next lab cycle.
M4 Notifications (REQ-13 to REQ-16)	Out of scope for current execution; SMS gateway and email server testing are planned later.
Load/stress testing above 100 users	No load testing tool configured in this lab.
Cross-browser testing	Chrome is selected as the primary browser for this cycle.
Mobile/responsive testing	LMS Version 1.0 is desktop-focused.
API security penetration testing	Requires specialized security tooling and separate security testing session.

6.6 Test Approach

Aspect	Approach
Testing Level	System testing of the integrated LMS application.
Testing Type	Functional black-box testing with selected performance checks.
Execution Method	Manual test execution through Jira test cases.
Black-box Techniques	Equivalence Partitioning and Boundary Value Analysis.
Priority Order	High priority test cases first, then Medium and Low.
Defect Management	Defects logged in Jira using severity and priority fields.
Re-testing	Fixed defects retested within 24 hours of deployment.
Regression Testing	Smoke tests TC-001 to TC-005 executed after major fixes.
Test Data	LMS-TESTDATA-001 restored before test sessions.

6.7 Pass / Fail Criteria

Level	Criteria
Test Case	Pass if actual result exactly matches expected result; fail for any deviation.
Test Cycle	Pass if all High and Medium priority test cases are executed and no Critical/High defects remain open.
Critical Defect	Testing stops immediately if login or core system access is completely blocked.

High Defect	Major functionality fails with no workaround.
Medium Defect	Function has a workaround but still needs correction.
Low Defect	Minor UI/cosmetic issue that does not block testing.

6.8 Suspension Criteria and Resumption Requirements

Suspension Criterion	Resumption Requirement
Critical login defect	Suspend all tests until fix is deployed and smoke tested.
More than 20% high severity failures	Pause execution and conduct defect triage.
Staging server unavailable over 2 hours	Suspend testing and contact DevOps.
Test data corruption	Restore test database snapshot before resuming.
Smoke test failure over 30%	Reject build and request a new build.

6.9 Environmental Needs

Need	Details
Browser	Google Chrome v120 or later
Operating System	Windows 10/11
Application URL	https://lms-staging.uni.edu
Database	PostgreSQL v14.2 test database
Accounts	Student and Library Staff test accounts
Network	Stable internet connection
Tools	Jira Cloud, Microsoft Word, Microsoft Excel

6.10 Responsibilities

Role	Responsibility
Student Tester	Create RTM, test plan, Jira test cases, and execute tests.
Instructor	Review lab deliverables and evaluate according to rubric.
Developer Role	Fix defects reported during testing.
DevOps Role	Maintain staging environment and test database.

6.11 Schedule

Activity	Description
Requirements Analysis	Prepare RTM skeleton and identify testable requirements.
Test Planning	Prepare IEEE 829 test plan.

Test Case Design	Create Jira test cases and link to requirements.
Environment Setup	Verify LMS staging environment and test data.
Execution	Run selected test cases in Jira.
Closure	Prepare submission report and checklist.

7. Jira Test Cases

TC-001 - Valid login with correct student credentials

Priority	High
Test Type	Functional
Preconditions	Student account exists and is not locked.
Test Steps	1. Open login page. 2. Enter valid email. 3. Enter valid password. 4. Click Login.
Expected Result	User is redirected to dashboard and welcome message is displayed.
Requirement Covered	REQ-01
Actual Result	To be filled during execution
Status	Not Executed

TC-002 - Invalid username login

Priority	High
Test Type	Functional
Preconditions	Login page is open.
Test Steps	1. Enter unregistered email. 2. Enter any password. 3. Click Login.
Expected Result	System displays invalid credentials message and does not log in.
Requirement Covered	REQ-01
Actual Result	To be filled during execution
Status	Not Executed

TC-003 - Account lockout after 3 failed attempts

Priority	High
Test Type	Functional, Security
Preconditions	Student account exists and is active.
Test Steps	1. Enter wrong password three times. 2. Try logging in again.
Expected Result	Account is locked after third failed attempt and lockout message is shown.
Requirement Covered	REQ-02

Actual Result	To be filled during execution
Status	Not Executed

TC-004 - Password reset email delivery

Priority	Medium
Test Type	Functional, Performance
Preconditions	Registered email account exists.
Test Steps	1. Open Forgot Password. 2. Enter registered email. 3. Submit request.
Expected Result	Password reset email is delivered within 2 minutes.
Requirement Covered	REQ-03
Actual Result	To be filled during execution
Status	Not Executed

TC-005 - Session timeout after 30 minutes inactivity

Priority	High
Test Type	Functional, Security
Preconditions	User is logged in.
Test Steps	1. Stay inactive for 30 minutes. 2. Try to access dashboard.
Expected Result	Session expires and user is redirected to login page.
Requirement Covered	REQ-04
Actual Result	To be filled during execution
Status	Not Executed

TC-006 - Search book by title

Priority	High
Test Type	Functional
Preconditions	Known book title exists in database.
Test Steps	1. Open search page. 2. Enter book title. 3. Click Search.
Expected Result	Matching book title appears in search results.
Requirement Covered	REQ-05
Actual Result	To be filled during execution
Status	Not Executed

TC-007 - Filter books by genre

Priority	Medium
Test Type	Functional
Preconditions	Books exist in multiple genres.
Test Steps	1. Search books. 2. Select a genre filter.
Expected Result	Only books from selected genre are displayed.
Requirement Covered	REQ-06
Actual Result	To be filled during execution
Status	Not Executed

TC-008 - Pagination with more than 20 results

Priority	Low
Test Type	Functional
Preconditions	More than 20 matching records exist.
Test Steps	1. Search broad keyword. 2. Observe results. 3. Click next page.
Expected Result	20 results appear per page and navigation controls work.
Requirement Covered	REQ-08
Actual Result	To be filled during execution
Status	Not Executed

8. Defect Report Template

Field	Description
Defect ID	BUG-001
Title	Short and clear defect title
Module	Authentication / Search / Loan / Notification
Severity	Critical / High / Medium / Low
Priority	High / Medium / Low
Steps to Reproduce	Numbered steps to reproduce the bug
Expected Result	What should happen
Actual Result	What actually happened
Environment	Browser, OS, LMS version
Attachment	Screenshot or screen recording if available

10. Conclusion

Lab Session 3 work has been completed for the University Library Management System. The report includes Jira setup details, a complete RTM with 100% coverage, IEEE 829 test plan sections, Jira test cases, and a submission checklist. This document can be submitted as a single Word file after pasting the actual Jira project URL and confirming instructor viewer access.

THE END